

Optimization I – Homework 1

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1. Let $f_1(x_1, x_2) = x_1^2 - x_2^2$, $f_2(x_1, x_2) = 2x_1x_2$. Represent the level sets associated with $f_1(x_1, x_2) = 12$ and $f_2(x_1, x_2) = 16$ on the same figure using Python. Indicate on the figure, the points $\mathbf{x} = [x_1, x_2]^\top$ for which $f(\mathbf{x}) = [f_1(x_1, x_2), f_2(x_1, x_2)]^\top = [12, 16]^\top$.

We may analytically solve the system

$$\begin{cases} x_1^2 - x_2^2 = 12 \\ 2x_1x_2 = 16 \end{cases} \quad (1)$$

to find that the intersection points are $(4, 2)$ and $(-4, -2)$.

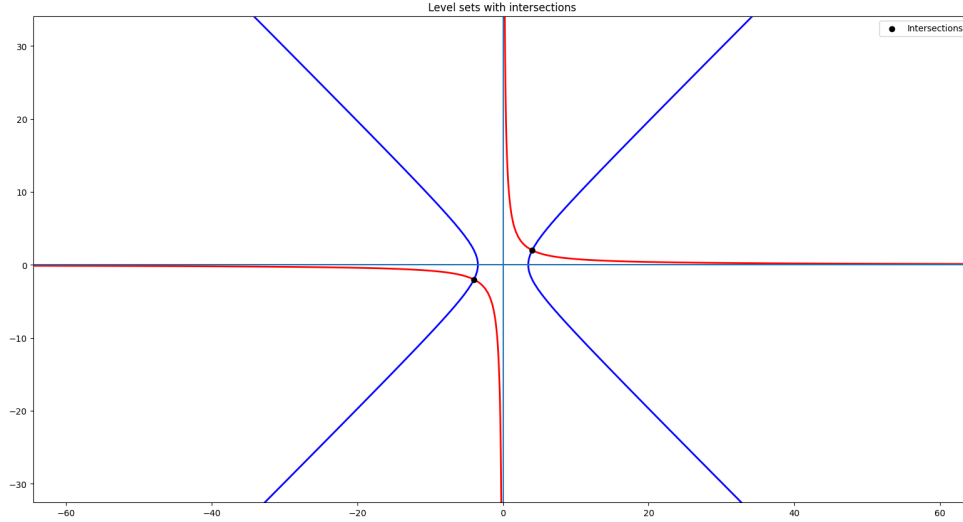


Figure 1: In blue, $x_1^2 - x_2^2 = 12$ and in red, $2x_1x_2 = 16$. The intersection points are given by $\{(4, 2), (-4, -2)\}$.

2. Consider the function $f(\mathbf{x}) = (\mathbf{a}^\top \mathbf{x})(\mathbf{b}^\top \mathbf{x})$, where $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{x}$ are n -dimensional vectors. Compute the gradient $\nabla f(\mathbf{x})$ and the Hessian $\nabla^2 f(\mathbf{x})$.

Proof. We define

$$h(\mathbf{x}) = \mathbf{a}^\top \mathbf{x}, \quad g(\mathbf{x}) = \mathbf{b}^\top \mathbf{x}. \quad (2)$$

Their gradients are given by

$$\nabla h(\mathbf{x}) = \mathbf{a}, \quad \nabla g(\mathbf{x}) = \mathbf{b}. \quad (3)$$

Now, by [Theorem 0.1](#) (Product Rule), we obtain

$$\begin{aligned}
 \nabla f(\mathbf{x}) &= \nabla(h(\mathbf{x})g(\mathbf{x})) \\
 &= g(\mathbf{x})\nabla h(\mathbf{x}) + h(\mathbf{x})\nabla g(\mathbf{x}) \\
 &= (\mathbf{b}^\top \mathbf{x})\mathbf{a} + (\mathbf{a}^\top \mathbf{x})\mathbf{b} \\
 (\mathbf{b}^\top \mathbf{x}, \mathbf{a}^\top \mathbf{x} \in \mathbb{R}) &= \mathbf{a}\mathbf{b}^\top \mathbf{x} + \mathbf{b}\mathbf{a}^\top \mathbf{x} \\
 &= (\mathbf{a}\mathbf{b}^\top + \mathbf{b}\mathbf{a}^\top)\mathbf{x}.
 \end{aligned} \tag{4}$$

Therefore, for the Hessian, we have

$$\begin{aligned}
 \nabla^2 f(\mathbf{x}) &= \nabla((\mathbf{a}\mathbf{b}^\top + \mathbf{b}\mathbf{a}^\top)\mathbf{x}) \\
 &= (\mathbf{a}\mathbf{b}^\top + \mathbf{b}\mathbf{a}^\top)^\top \\
 &= \mathbf{a}\mathbf{b}^\top + \mathbf{b}\mathbf{a}^\top.
 \end{aligned} \tag{5}$$

□

3. Compute the gradient of

$$f(\theta) \stackrel{\text{def}}{=} \frac{1}{2} \sum_{i=1}^n [g(\mathbf{x}_i) - g(\mathbf{A}\mathbf{x}_i + \mathbf{b})]^2 \tag{6}$$

with respect to θ , where $\theta = [a_{11}, a_{12}, a_{21}, a_{22}, b_1, b_2]^\top$, $\mathbf{x}_i \in \mathbb{R}^2$, $\mathbf{A} \in \mathbb{R}^{2 \times 2}$, $\mathbf{b} \in \mathbb{R}^2$ are defined as follows:

$$\begin{aligned}
 \mathbf{A} &= \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \\
 \mathbf{b} &= [b_1, b_2]^\top.
 \end{aligned} \tag{7}$$

and $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ is $\mathcal{C}^1$.

Proof. Define

$$\varepsilon_{i(\theta)} = g(\mathbf{x}_i) - g(\mathbf{A}\mathbf{x}_i + \mathbf{b}). \tag{8}$$

Thus, we write

$$f(\theta) = \frac{1}{2} \sum_i \varepsilon_i(\theta)^2. \tag{9}$$

We have that

$$\nabla_{\theta} \varepsilon_i^2(\theta) = 2\varepsilon_i(\theta) \nabla_{\theta} \varepsilon_i(\theta). \tag{10}$$

Therefore,

$$\nabla f(\theta) = \sum_i \varepsilon_i(\theta) \nabla_{\theta} \varepsilon_i(\theta). \tag{11}$$

Now note that

$$\begin{aligned}
 \nabla_{\theta} \varepsilon_i(\theta) &= \nabla_{\theta} [g(\mathbf{x}_i) - g(\mathbf{Ax}_i + \mathbf{b})] \\
 &= -\nabla g(\mathbf{Ax}_i + \mathbf{b}) \nabla_{\theta} (\mathbf{Ax}_i + \mathbf{b}) \\
 &= -\nabla g(\mathbf{Ax}_i + \mathbf{b}) [e_1 \mathbf{x}_i, e_2 \mathbf{x}_i, e_3 \mathbf{x}_i, e_4 \mathbf{x}_i, e_5, e_6]^{\top} \\
 &= \begin{bmatrix} -\nabla g(\mathbf{Ax}_i + \mathbf{b}) \mathbf{x}_i^{\top} \\ -\nabla g(\mathbf{Ax}_i + \mathbf{b}) \end{bmatrix}.
 \end{aligned} \tag{12}$$

Therefore, we conclude

$$\nabla f(\theta) = \sum_i (g(\mathbf{x}_i) - g(\mathbf{Ax}_i + \mathbf{b})) \begin{bmatrix} -\nabla g(\mathbf{Ax}_i + \mathbf{b}) \mathbf{x}_i^{\top} \\ -\nabla g(\mathbf{Ax}_i + \mathbf{b}) \end{bmatrix}. \tag{13}$$

□

4. Let $f(r, \theta)$ be a function from $\mathbb{R}^2$ to $\mathbb{R}$ with $r = \sqrt{x^2 + y^2}$ and $\theta = \arctan \frac{y}{x}$. Compute $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

Proof. We first compute the partial derivatives of $r(x, y)$ and $\theta(x, y)$. These are given by

$$\begin{aligned}
 \frac{\partial r}{\partial x} &= \frac{x}{\sqrt{x^2 + y^2}} = \frac{x}{r}, & \frac{\partial r}{\partial y} &= \frac{y}{\sqrt{x^2 + y^2}} = \frac{y}{r}. \\
 \frac{\partial \theta}{\partial x} &= -\frac{\frac{y}{x^2}}{1 + \frac{y^2}{x^2}} = -\frac{y}{x^2 + y^2} = -\frac{y}{r^2}, & \frac{\partial \theta}{\partial y} &= \frac{\frac{1}{x}}{\sqrt{1 + \frac{y^2}{x^2}}} = \frac{x}{\sqrt{x^2 + y^2}} = \frac{x}{r^2}.
 \end{aligned} \tag{14}$$

Therefore, by the chain rule,

$$\begin{aligned}
 \frac{\partial f}{\partial x} &= \frac{\partial f}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial f}{\partial \theta} \frac{\partial \theta}{\partial x} = \frac{\partial f}{\partial r} \frac{x}{r} - \frac{\partial f}{\partial \theta} \frac{y}{r^2}, \\
 \frac{\partial f}{\partial y} &= \frac{\partial f}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial f}{\partial \theta} \frac{\partial \theta}{\partial y} = \frac{\partial f}{\partial r} \frac{y}{r} + \frac{\partial f}{\partial \theta} \frac{x}{r^2}.
 \end{aligned} \tag{15}$$

□

5. The directional derivatives $\frac{\partial f}{\partial v}(x_0, y_0, z_0)$ of a differentiable function f are $\frac{3}{\sqrt{2}}, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}$ in the directions of the vectors $\begin{bmatrix} 0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \end{bmatrix}^{\top}$, $\begin{bmatrix} \frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \end{bmatrix}^{\top}$ and $\begin{bmatrix} \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \end{bmatrix}^{\top}$. Compute $\nabla f(x_0, y_0, z_0)$.

Proof. Note that the three directions form a basis for $\mathbb{R}^3$, and thus the gradient can be completely determined. Given that f is $\mathcal{C}^1$ and each v is a unit vector,

$$\nabla f(x_0, y_0, z_0) \cdot v = D_v f(x_0, y_0, z_0), \tag{16}$$

Let $\nabla f(x_0, y_0, z_0) = (x, y, z)$, then we have a system of three equations:

$$\begin{aligned}
 0x + \frac{y}{\sqrt{2}} + \frac{z}{\sqrt{2}} &= \frac{3}{\sqrt{2}} \\
 \frac{x}{\sqrt{2}} + 0y + \frac{z}{\sqrt{2}} &= \frac{1}{\sqrt{2}} \\
 \frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}} + 0z &= -\frac{1}{\sqrt{2}}
 \end{aligned} \tag{17}$$

which simplifies to

$$\begin{aligned} y + z &= 3 \\ x + z &= 1 \\ x + y &= -1 \end{aligned} \tag{18}$$

and thus

$$\nabla f(x_0, y_0, z_0) = \left(-\frac{3}{2}, \frac{1}{2}, \frac{5}{2}\right). \tag{19}$$

□

6. Show that the level curves of the function $f(x, y) = x^2 + y^2$ are orthogonal to the level curves of $g(x, y) = \frac{y}{x}$ for all (x, y) .

Proof. The level curves of f at $l \in \mathbb{R}$ are given by

$$S_f = \{(x, y) : x^2 + y^2 = l\}. \tag{20}$$

Similarly,

$$S_g = \left\{(x, y) : \frac{y}{x} = l\right\}. \tag{21}$$

That is, we must verify if the curves

$$y^2 = l - x^2 \quad y = lx \tag{22}$$

are orthogonal. Their derivatives (slopes) are

$$m_1 = \frac{dy}{dx} = -\frac{x}{y}, \quad m_2 = \frac{dy}{dx} = l \tag{23}$$

and thus their product is

$$m_1 m_2 = -\frac{lx}{y}. \tag{24}$$

Substituting $y = lx$, we have

$$m_1 m_2 = -1. \tag{25}$$

That is, the curves are orthogonal. □

7. Let f, g, h be differentiable functions, with $f : \mathbb{R}^n \rightarrow \mathbb{R}^3$, $g : \mathbb{R}^n \rightarrow \mathbb{R}^3$ and $h : \mathbb{R}^n \rightarrow \mathbb{R}$

$$h(\mathbf{x}) = f(\mathbf{x})^\top g(\mathbf{x}) \tag{26}$$

show that

$$Dh(\mathbf{x}) = f(\mathbf{x})^\top Dg(\mathbf{x}) + g(\mathbf{x})^\top Df(\mathbf{x}). \tag{27}$$

Proof. We have that

$$\lim_{\|k\| \rightarrow 0} \frac{|h(a+k) - h(a) - Dh(a)k|}{\|k\|} = \lim_{\|k\| \rightarrow 0} \frac{|f(a+k)g(a+k) - f(a)g(a) - Df(a)g(a)k|}{\|k\|} \tag{28}$$

Now, note that

$$\begin{aligned} f(a+k)g(a+k) - f(a)g(a) &= f(a+k)g(a+k) + f(a)g(a+k) - f(a)g(a+k) - f(a)g(a) \\ &= f(a)[g(a+k) - g(a)] + g(a+k)[f(a+k) - f(a)]. \end{aligned} \quad (29)$$

Thus, (28) is equivalent to

$$\lim_{\|k\| \rightarrow 0} \frac{|f(a)[g(a+k) - g(a)] + g(a+k)[f(a+k) - f(a)] - Df(a)g(a)k|}{\|k\|} \quad (30)$$

Since both f, g are differentiable, we have

$$\lim_{\|k\| \rightarrow 0} \frac{|f(a)Dg(a) + g(a)Df(a) - Df(a)g(a)k|}{\|k\|} \quad (31)$$

□

8. Consider the **induced matrix norm**

$$\|A\|_p = \max_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p} \quad (32)$$

where $\|\cdot\|_p$ is the ℓ_p norm, i.e.

$$\|x\|_p = (|x_1|^p + |x_2|^p + \dots + |x_n|^p)^{\frac{1}{p}} \quad (33)$$

Show that

$$\|AB\|_p \leq \|A\|_p \|B\|_p. \quad (34)$$

Proof. We first note that by definition, $\frac{\|Ax\|_p}{\|x\|_p} \leq \|A\|_p$ for all $x \neq 0$. Therefore,

$$\|Ax\|_p \leq \|A\|_p \|x\|_p, \quad x \neq 0. \quad (35)$$

Thus,

$$\|ABx\|_p = \|A(Bx)\|_p \leq \|A\|_p \|Bx\|_p \leq \|A\|_p \|B\|_p \|x\|_p. \quad (36)$$

Equivalently,

$$\frac{\|ABx\|_p}{\|x\|_p} \leq \|A\|_p \|B\|_p, \quad x \neq 0. \quad (37)$$

Since $x \in \mathbb{R}^n \setminus \{0\}$ was arbitrary, we may conclude that

$$\|AB\|_p = \max_{x \neq 0} \frac{\|ABx\|_p}{\|x\|_p} \leq \|A\|_p \|B\|_p. \quad (38)$$

□

Appendix

Theorem 0.1. (Product Rule) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be two differentiable functions. Define the function $h : \mathbb{R}^n \rightarrow \mathbb{R}$ by $h(x) = f(x)^\top g(x)$ then

$$D_x h(x) = f(x)^\top Dg(x) + g(x)^\top Df(x) \in \mathbb{R}^{1 \times n}, \quad (39)$$

where $Dg(x)$ and $Df(x)$ are matrices (the Jacobian matrix). Equivalently,

$$\nabla h(\mathbf{x}) = \nabla \mathbf{f}(\mathbf{x})\mathbf{g}(\mathbf{x}) + \nabla \mathbf{g}(\mathbf{x})\mathbf{f}(\mathbf{x}) \in \mathbb{R}^n. \quad (40)$$